

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular Winter Examination – 2024

Course: B. Tech

Branch: Common To All Branches

Semester: I

Subject Code & Name: 24AF1EMES104 & Engineering Mechanics

Max Marks: 60

Date: 11/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

(Level Marks
/CO)

Q. 1 Objective type questions. (Compulsory Question)

12

- | | | | |
|---|---|----|---|
| 1 | Which of the following is an example of a concentrated load? | L1 | 1 |
| | a) Self-weight of a beam | | |
| | b) Wind load | | |
| | c) Point force applied at a node | | |
| | d) Hydrostatic pressure | | |
| 2 | In a free body diagram, the weight of the object is usually represented as acting: | L1 | 1 |
| | a) Horizontally | | |
| | b) At the centroid vertically downward | | |
| | c) At the support points | | |
| | d) Tangentially | | |
| 3 | A force couple consists of: | L1 | 1 |
| | a) Two equal and opposite forces acting along the same line | | |
| | b) Two equal and opposite forces with a separation distance | | |
| | c) A single force acting at a point | | |
| | d) Three non-parallel forces | | |
| 4 | If a force system is in equilibrium, the algebraic sum of the moments about any point is: | L1 | 1 |
| | a) Maximum | | |
| | b) Minimum | | |
| | c) Zero | | |
| | d) Constant | | |
| 5 | The analytical conditions for equilibrium in two-dimensional force systems are: | L1 | 1 |
| | a) $\Sigma F_x = 0$ and $\Sigma F_y = 0$ | | |
| | b) $\Sigma F_x = 0$, $\Sigma F_y = 0$, and $\Sigma M_z = 0$ | | |
| | c) $\Sigma F_x \neq 0$, $\Sigma F_y = 0$ | | |
| | d) $\Sigma M_x = 0$, $\Sigma M_y = 0$ | | |

6	The centroid of a rectangle is located at:	L1	1
	a) One-fourth of the height and one-fourth of the width		
	b) At the intersection of diagonals		
	c) At the midpoint of any side		
	d) One-third of the height from the base		
7	In a Hinged or Pinned Support, how many reactions are developed?	L1	1
	a) Two		
	b) Three		
	c) Four		
	d) Six		
8	The equation used to determine if a truss is perfect:	L1	1
	a) $m+4=2j$		
	b) $m=2j-3$		
	c) $2m=3+j$		
	d) $m=j+r$		
9	If an object is thrown upward with an initial velocity u , the velocity at the highest point is:	L1	1
	a) Equal to u		
	b) Zero		
	c) Infinite		
	d) Equal to acceleration		
10	The area under the acceleration-time graph gives:	L1	1
	a) Velocity		
	b) Displacement		
	c) Momentum		
	d) Force		
11	The kinetic energy of a rigid body in pure translation is given by:	L1	1
	a) $\frac{1}{2} mv^2$		
	b) mv^2		
	c) mgh		
	d) $\frac{1}{2} I\omega^2$		
12	The coefficient of restitution for a perfectly elastic collision is:	L1	1
	a) 0		
	b) 1		
	c) Greater than 1		
	d) Between 0 and 1		
Q. 2	Solve the following.		12
A)	The following forces act at a point:	L2	6
	(i) 20 N inclined at 30° towards North of East,		
	(ii) 25 N towards North,		
	(iii) 30 N towards North West, and		
	(iv) 35 N inclined at 40° towards South of West.		
	Find the magnitude and direction of the resultant force.		
B)	State and prove Varignon's theorem.	L1	6

Q.3 Solve the following.

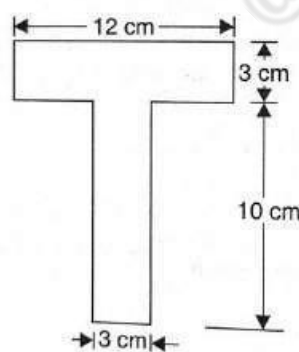
12

- A) Locate the centroid of the T section shown in

figure.

L2

6



- B) A body of weight 500 N is pulled up an inclined plane by a force of 350 N. The Inclination of the plane is 30° to the horizontal and the force is applied parallel to the plane. Determine the coefficient of friction.

L2

6

Q. 4 Solve Any Two of the following.

12

- A) What are the types of loads and explain them in details with neat sketches?

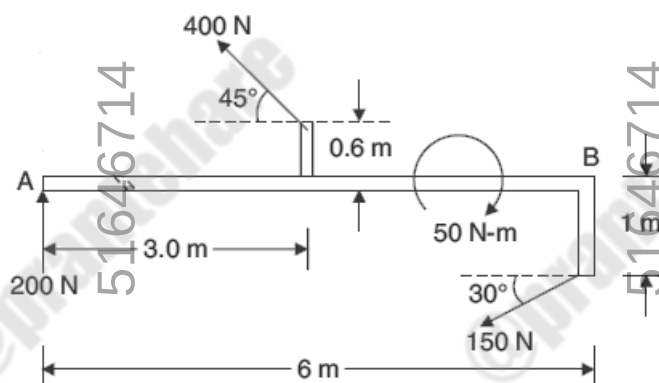
L1

6

- B) A bracket is subjected to three forces and a couple as shown in figure. Determine magnitude, direction and the line of action of the resultant.

L3

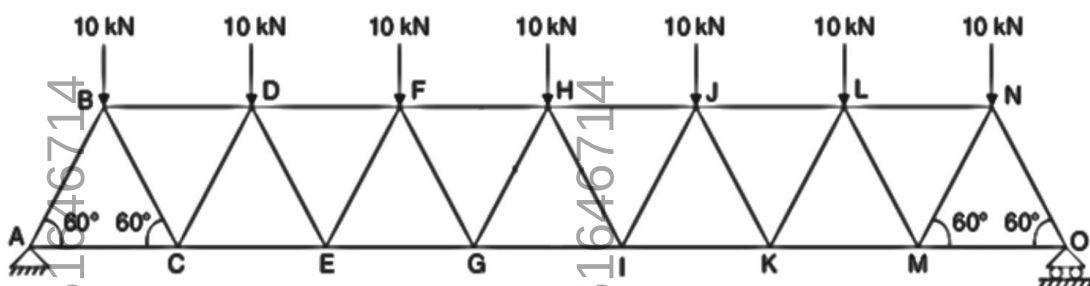
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- C) Determine the forces in the members FH, HG and GI in the truss shown in figure. Each load is 10 kN and all triangles are equilaterals with sides equal to 4 m.

L3

6



Q.5 Solve Any Two of the following.

12

- A) Prove equations of motion of a body moving with constant acceleration.

L2

6

- B) The motion of a particle moving in a straight line is given by the expression

L3

6

$$s = t^3 - 3t^2 + 2t + 5$$

where, s is the displacement in metres and t is the time in seconds.

Determine: (1) velocity and acceleration after 4 seconds;

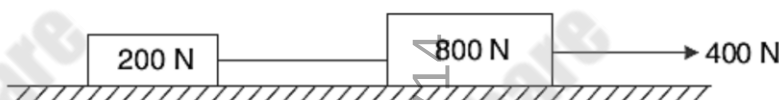
(2) maximum or minimum velocity and corresponding displacement;

(3) time at which velocity is zero.

- C) A ball is dropped from the top of a tower 30 m high. At the same instant a second ball is thrown upward from the ground with an initial velocity of 15 m/sec. When and where do they cross and with what relative velocity? L1 6

Q. 6 Solve Any Two of the following. 12

- A) Two weights 800 N and 200 N are connected by a thread and they move along a rough horizontal plane under the action of a force of 400 N applied to the 800 N weight as shown in figure. The coefficient of friction between the sliding surface of the weights and the plane is 0.3. Using D' Alembert's principle, determine the acceleration of the weight and tension in the thread. L2 6



- B) State and prove work energy principle. L1 6
- C) A 1500 N block is in contact with a level plane, the coefficient of friction between two contact surfaces being 0.1. If the block is acted upon by a horizontal force of 300 N, what time will elapse before the block reaches a velocity of 16 m/sec starting from rest? If 300 N force is then removed, how much longer will the block continue to move? Solve the problem using impulse momentum equation. L3 6

*** End ***